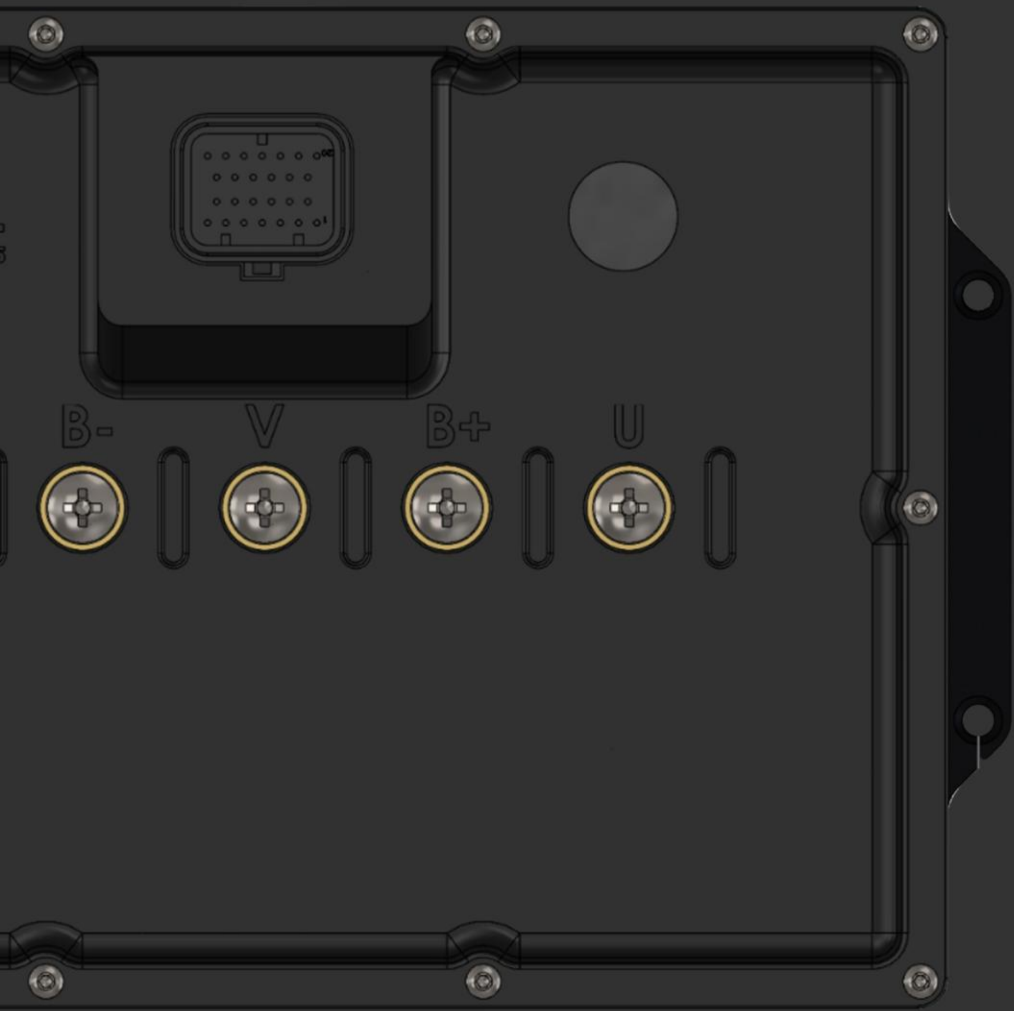


Performance for All

C1400/CL1400/CC1000
CONTROLLERS
User Manual

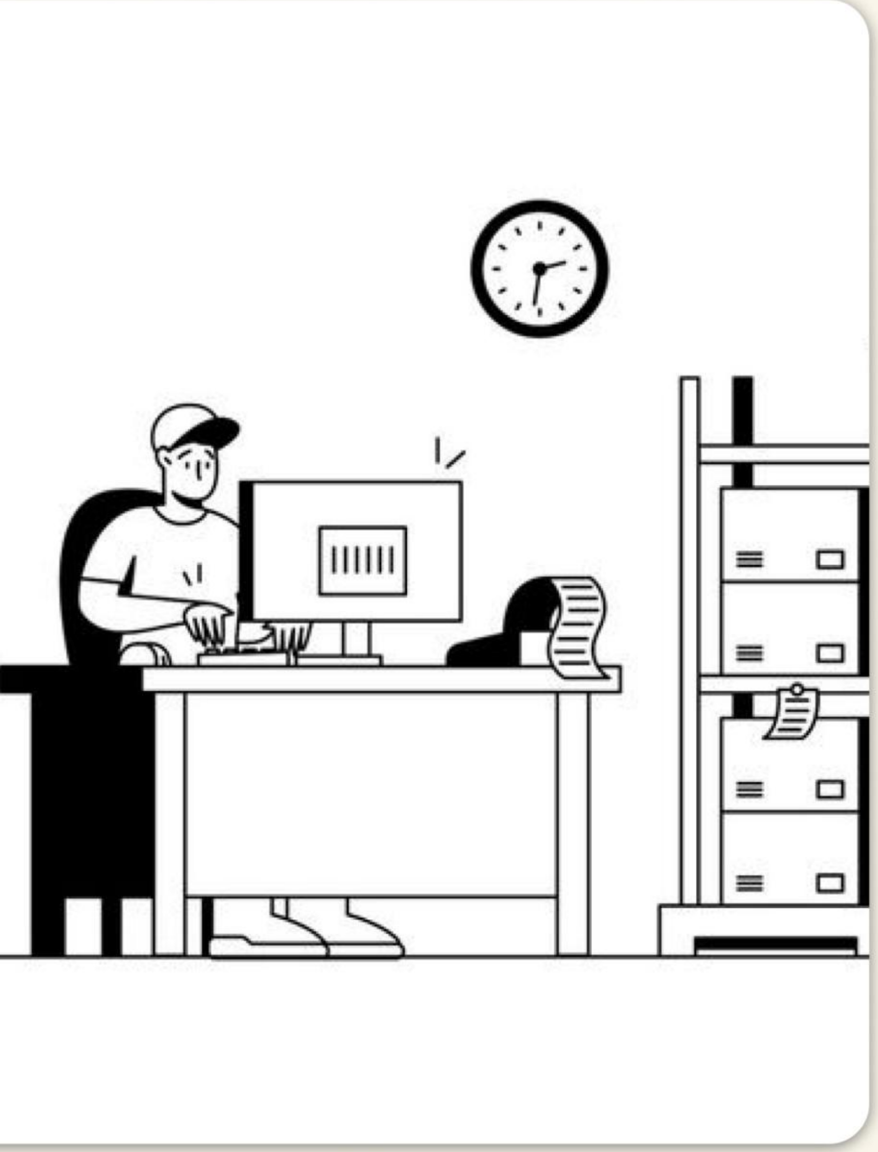
Overview

C1400/CL1400/CC1000 is a high-performance motor controller designed for EV applications like 4W, 2W, cars, bikes, and go-karts. It runs on open-source VESC software and is fully customizable. Backed by a skilled team with 6+ years of experience, it offers better performance than any other controller series globally. The product comes with no warranty, but manufacturing defects are covered (terms at 3shulmotors.com). A one-time free remote setup is included; further support is chargeable.

Warranty

Controller comes with warranty on the product which must be claimed by booking a remote setup session with our technical team at support@3shulmotors.com after receiving the product you bought. Only then the replacement warranty can be claimed in which; (1) If any major breakdown occurs in the controller product after setup by company or (2) If any manufacturing defect in the product (Term & Conditions apply – https://3shulmotors.com). If the controller is setup by your own side without claiming the warranty by remote setup with us then warranty will be void. Any damage occurs while use of products 3Shul Motors will not be responsible then. Further remote setup for troubleshooting service, consultation for upgrade and modifications will be chargeable on hourly basis.

Tech. Specs.

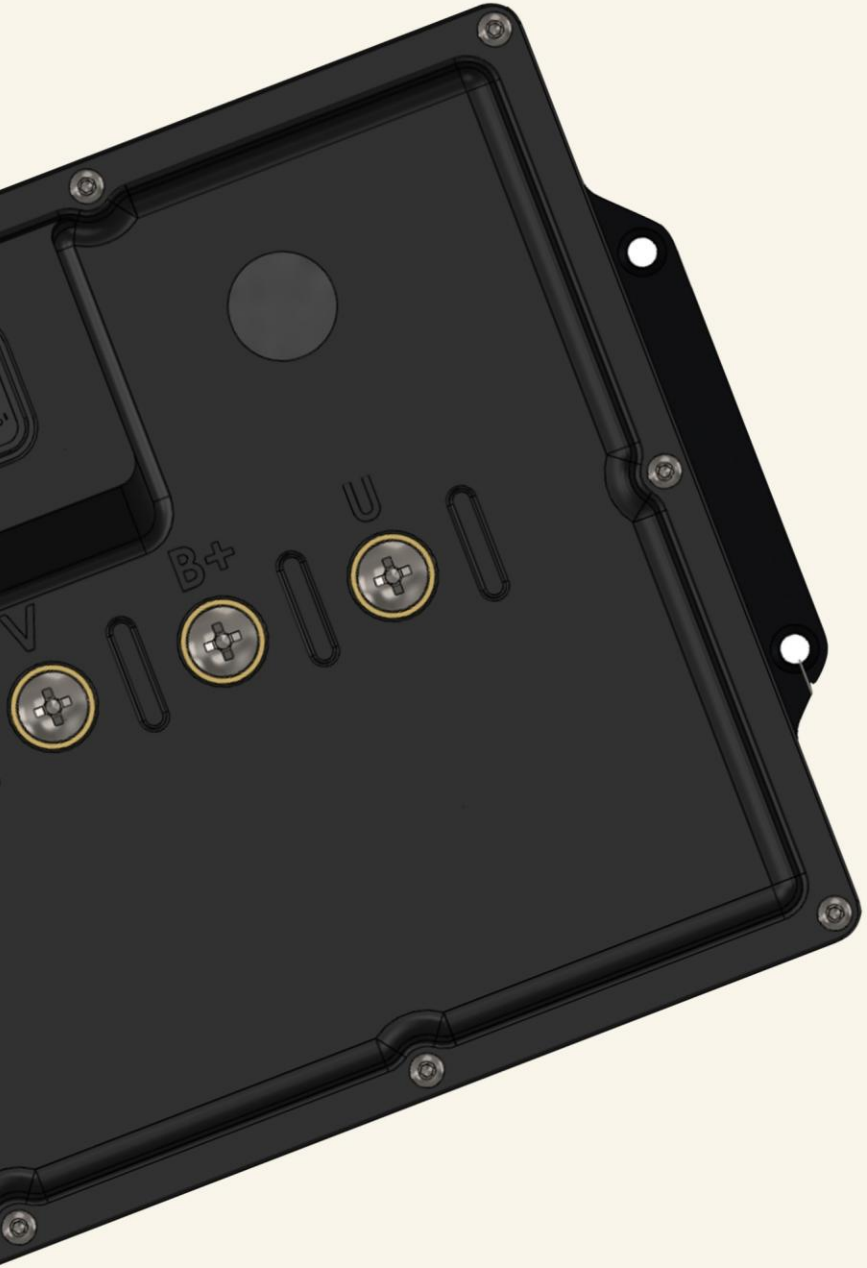


VOLTAGE AND CURRENT RATING	C1400: 36 - 92V (8-22s safe)/ 800A Battery, 1400A Phase CC1000: 36-144V (8-40S safe)/ 500A Battery, 1000A Phase (values depend on mounting and ambient temperature around the device) CL1400: 36 - 126V (8-30s safe)/ 800A Battery, 1400A Phase
MOTOR CONTROL MODES	BLDC SQUARE WAVE AND FOC SINE WAVE
FIRMWARE	LATEST (firmware update supported)
ERPM	100000 (tested)
CONTROL INTERFACE PORTS	USB, CAN, UART, BLUETOOTH
SUPPORTED SENSORS	ABI, HALL, AS5047, AS5048A, SENSOR-LESS, SIN-COS
INPUT SET SUPPORT	PPM, ADC, NRF, UART
Bus Bar Diameter	12 mm
PROGRAMMABLE	YES
MAX POWER OUTPUT	C700: 25 KW, CL700: 35 KW C1000: 40 KW, CL1000: 50 KW
MAX ZERO VECTOR FREQUENCY	30 KHz

Features

- Field Weakening and MTPA (Maximum Torque Per Amp) supported
- IMU (Gyroscope and Accelerometer) Onboard (BMI160)
- 12V 5Amp DC-DC converter for powering accessories
- 120mm fan mount for forced cooling
- Plenty of CPU resources left for custom apps on the controller
- Powered by STM32F4 microcontroller running at 168MHz
- Sensor and sensor-less FOC with auto-detection of all motor parameters
- Current and voltage measurement on all phases
- Variable regenerative braking
- Supports Brushless, Brushed DC, and IPM motors
- GUI that is easy on the eyes
- Adaptive PWM frequency for better ADC measurements
- Good start-up torque in both sensor-less and sensored mode
- Motor acts as a tachometer (useful for odometry)
- Duty-cycle Control, Speed Control, Current Control, and Position Control
- Seamless 4-quadrant operation
- Interface to control the motor: PPM signal (RC servo), analog, UART, I2C, USB, or CAN-bus
- Consumed and regenerated amp-hour and watt-hour counting
- Optional PPM signal output
- USB port uses modem profile (Android device can be connected directly)
- Adjustable protection against:

1. Low input voltage limit
2. High input voltage limit
3. High motor current limit
4. High input current limit
5. High regenerative braking current limit (motor and input separate)
6. Rapid duty cycle changes (ramping)
7. High RPM limit (separate for each direction)

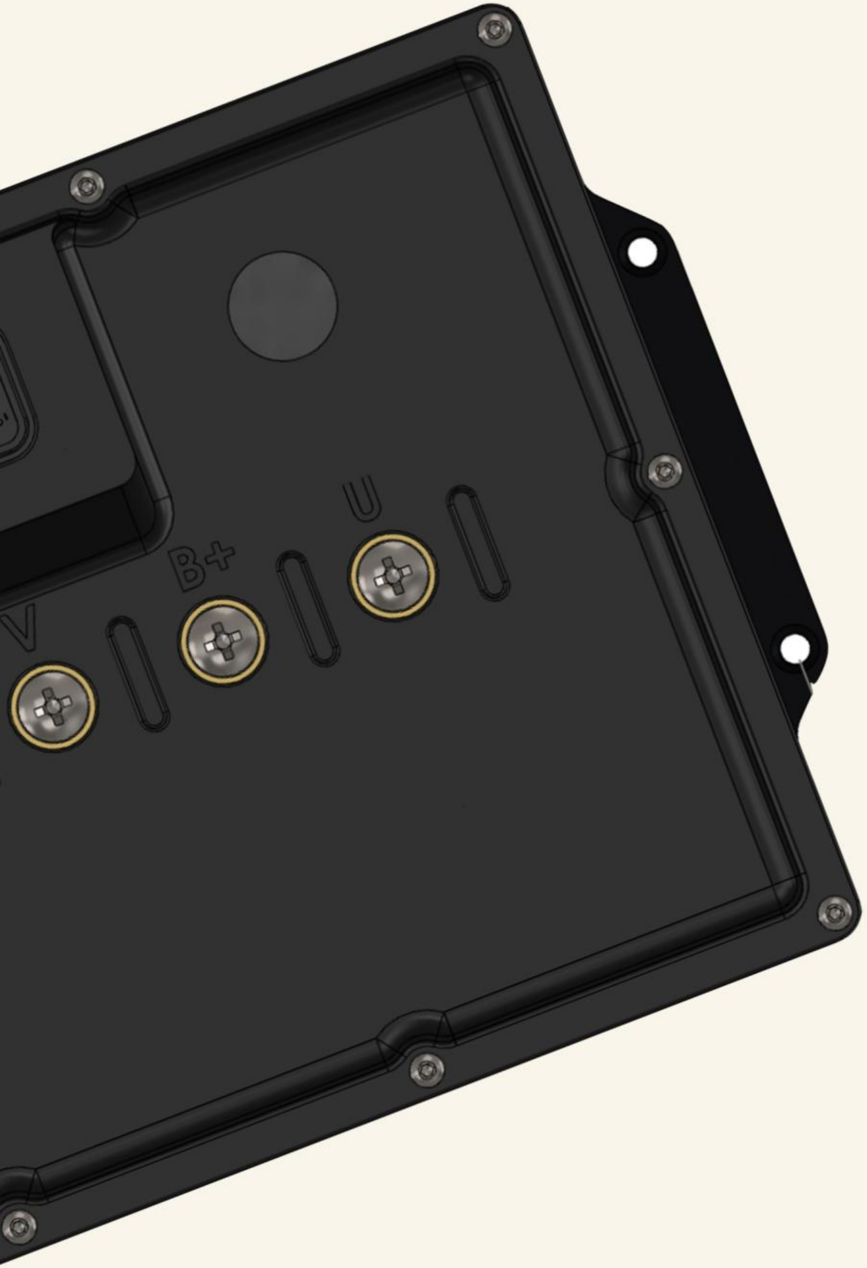


Design and Casing

The controller is enclosed in an aluminium and resin-printed enclosure. The aluminium part is positioned at the bottom, serving as a heat sink, while the resin-printed enclosure is specifically designed to cover the entire component aesthetically. The terminals are clearly visible and conveniently placed for easy operation. The overall structure is rigid, making it suitable for use in a variety of applications. Additionally, the compact size allows it to fit in limited spaces. The bottom of the aluminium heatsink features a finned surface and includes a 120mm mount for efficient cooling.

Hardware Protection

The hardware is equipped with multiple protection features including over-current protection, PWM overlap protection, short-circuit protection for power rails, and ESD-protected inputs. These features ensure the system operates reliably and is safeguarded against common electrical faults.



Hardware Setup

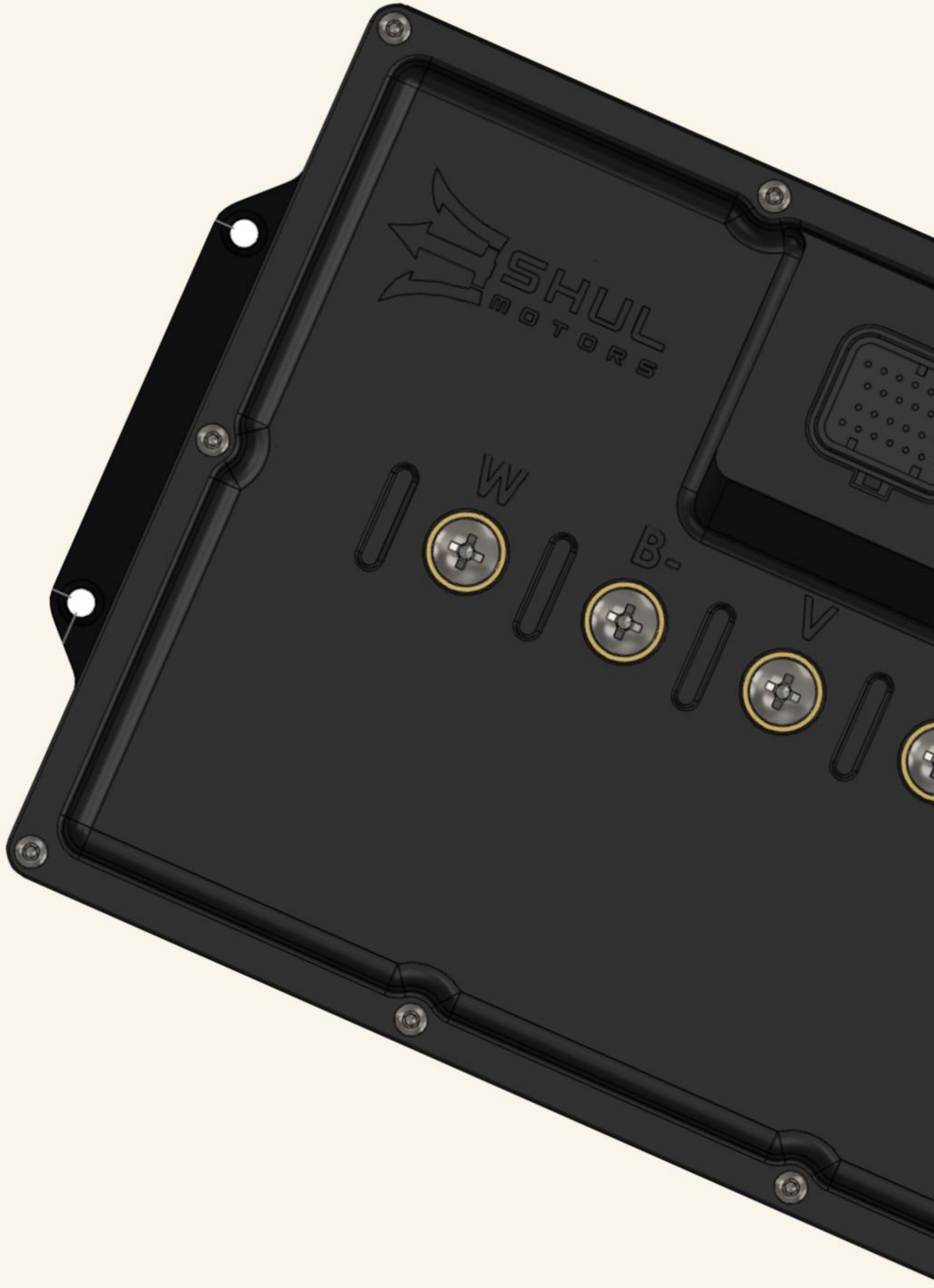
In the box, you will find the controller unit, a pack of terminal bolts for connections and mounting, a wiring harness with complete connectors (optional), and a manual/brochure with stickers. To begin setup, crimp lugs for U, V, W, and Battery +, - terminals. Start the assembly by fixing the controller on the table setup or in the vehicle test unit with the provided fitting accessories. Tighten the lug bolts of M6 × 1.0 with spring washer up to 8 Nm to ensure good connection or fitting. The wiring harness is included in the box; you can refer to the pin-out section in the manual for connections. A setup call can be arranged for the first-time setup of the controller for software configuration according to the application. After setup, enjoy the beast power output of the controller on your projects or vehicles. For any other problem faced during installation, you can email it at support@3shulmotors.com; our team will respond to your query as soon as possible. Check for any damage; if found, let us know through our website form at 3shulmotors.com or e-mail support@3shulmotors.com. Please read all the terms & conditions before purchasing the product from 3 Shul Motors for claiming warranties or making returns.

Software Setup

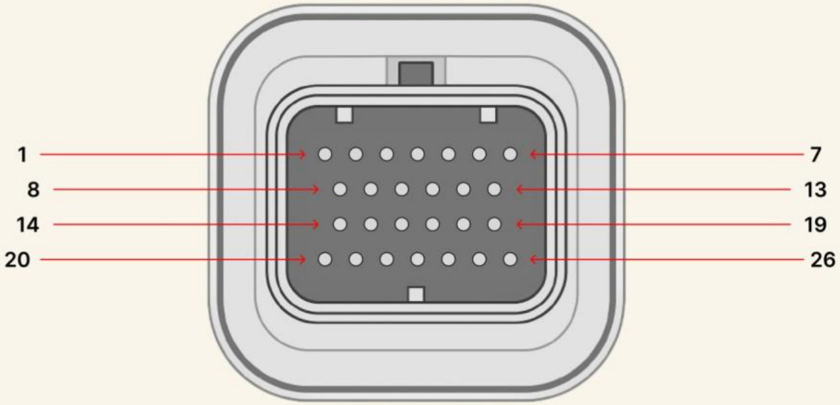
For the software setup of the controller, please follow the step-by-step guide available at the following link: https://vesc-project.com/node/178. If you encounter any issues while setting up the controller, feel free to contact us at support@3shulmotors.com for assistance.

Position Sensors

Multiple position sensor types are supported with this controller: Hall sensor, Encoder, SPI/Serial sensors such as AS5047P, AS5048D, TS5700N8501/Multiturn, and MT6816. It also supports ABI, Analogue SIN-COS encoders, and Sensorless configurations.



Bluetooth PIN for app connection
BT PIN: 147258



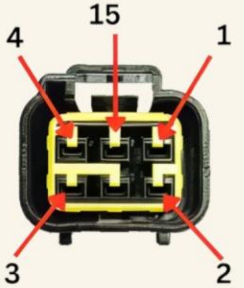
Connector

26 Pin Connector Pin Out

The Control Interface Mating Connector Is A 26-Pin AMPSEAL With The Following Ordering Codes: The Connector Has A Manufacturer Part Number Of 6437288-6, The Mating Housing Is 3-1437290-7, And The Female Crimp Pins Have A Part Number Of 3-1447221-3.

PIN NO.	DESCRIPTION	WIRE COLOR CODE	PIN NO.	DESCRIPTION	WIRE COLOR CODE
1.	5 VOLT	RED	14.	CAN_HIGH	BROWN
2.	SINE	PINK	15.	MOTOR_TEMP	WHITE
3.	COS	BLUE	16.	HALL_U	YELLOW
4.	GROUND (-)	BLACK	17.	HALL_V	GREEN
5.	TX (CRUISE)	YELLOW	18.	HALL_W	BLUE
6.	RX (REVERSE)	GREEN	19.	5 VOLT	RED
7.	5 VOLT	RED	20.	ADC1 THROTTLE INPUT	GREEN
8.	GROUND (-)	BLACK	21.	ADC2 REGEN INPUT	BLUE
9.	USB_D-	BROWN	22.	RC_PWM	WHITE
10.	USB_D+	PINK	23.	AUXOUT 12V GROUND 1	WHITE
11.	3.3 VOLT	N/A	24.	12 VOLT POSITIVE	RED
12.	MODE	N/A	25.	AUXOUT 12V GROUND 1	WHITE
13.	CAN_LOW	YELLOW	26.	VIN HIGH VOLTAGE	RED

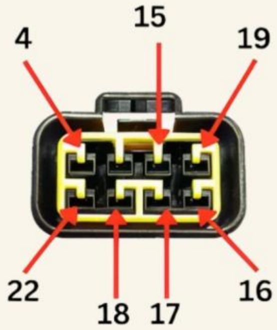
WIRING HARNESS CONNECTORS PIN OUT DESCRIPTION



SINE COS ENCODER COUPLER

PIN DESCRIPTION

1: +5V
15: MOTOR_TEMP
4: GROUND (-)
2: SINE
3: COS



HALL SENSOR COUPLER

PIN DESCRIPTION

19: +5v
15: MOTOR TEMP
4: GROUND (-)
16: HALL_U
17: HALL_V
18: HALL_W
22: RC_PWM

THROTTLE AND REGEN COUPLER

PIN DESCRIPTION

7: +5V
4: GROUND (-)
20: ADC1 (Throttle Input)
21: ADC2 (Regen Throttle
Input

AUXILARY SUPPLY

PIN DESCRIPTION

24: 12V+
24: 12V+
25: AUXOUT 12V
GROUND 2

CAN COUPLER

PIN DESCRIPTION

1: +5V
8: GROUND (-)
13: CAN_LOW
14: CAN_HIGH

USB COUPLER

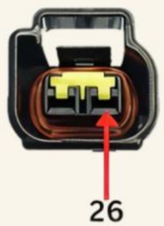
PIN DESCRIPTION

10: USB_D (+)
9: USB_D (-)
8: GROUND (-)

CRUISE AND REVERSE COUPLER

PIN DESCRIPTION

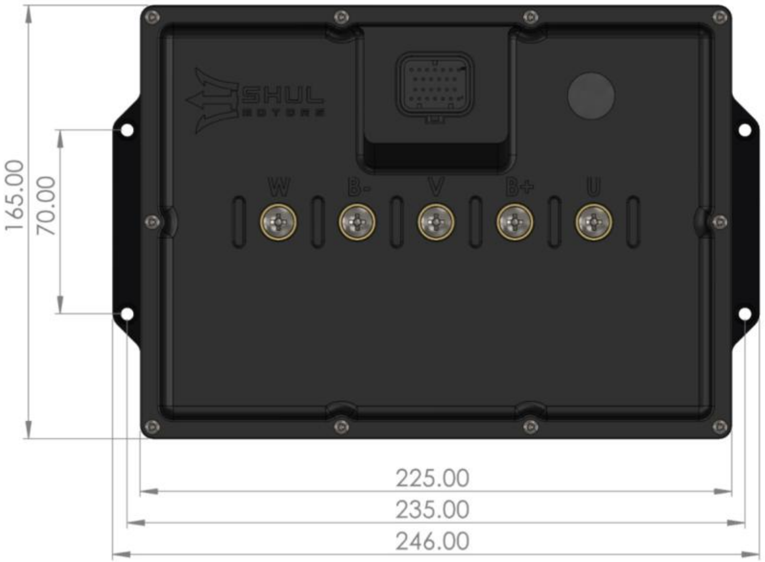
5: TX (CRUISE)
6: RX (REVERSE)
8: GROUND (-)



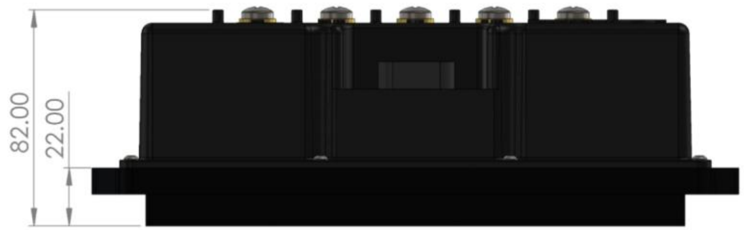
IGNITION INPUT COUPLER (VIN)

PIN DESCRIPTION

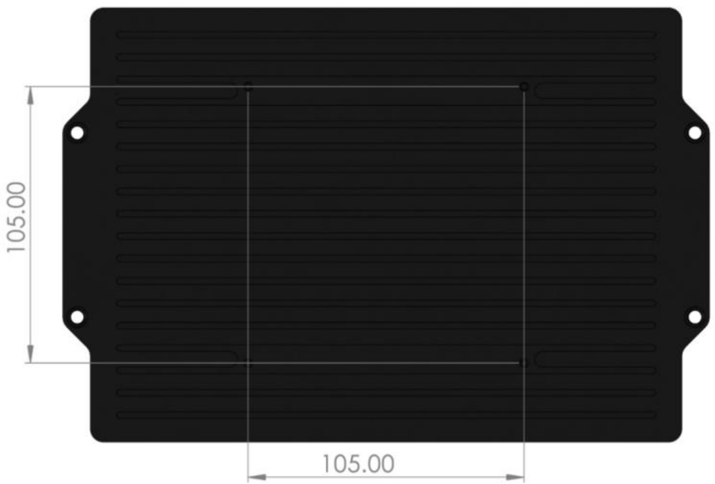
26: IGNITION INPUT
(HIGH VOLTAGE)



Top View



Side View



Bottom View

Our Factory

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Area, Bhilwara, Rajasthan - 311001

Office

402, Aarya Epoch, Vijay Cross Roads,
Navrangpura, Ahmedabad, GJ - 380009